Hidden dimensions: square roots and formula checks

Day 16 • 2026-11-26 • 2:15 PM–3:30 PM

15 questions. Show your method and include units. For multiple choice, enter A, B, C or D. Follow your instructor’s calculator guidance.

Name _____ **Date** _____

Type into the fields and save a copy, or print and write.

READ & REASON

01 For a square with area A, which operation finds the positive side length?

A. Divide A by 2

B. Square A

C. Multiply A by 4

D. Take the square root of A

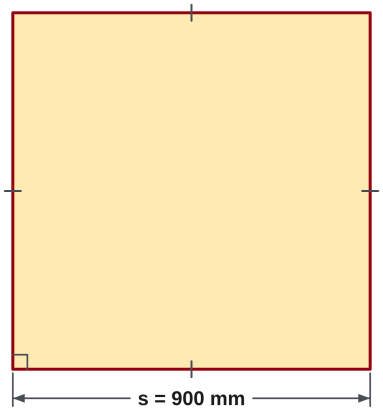
Choice (A–D) _____ **Working / reason** _____

Textbook sections 8.6 pp.201–203; sections 8.7 pp.203–204; calculator pp.51–53

Hidden dimensions: square roots and formula checks

Day 16 • 2026-11-26 • 15 questions

02 Use the 900 mm side shown by Shape Canvas to calculate the square’s area.



Shape Canvas

Answer (mm²)

Working / reason

Textbook sections 8.6 pp.201–203; sections 8.7 pp.203–204; calculator pp.51–53

Hidden dimensions: square roots and formula checks

Day 16 • 2026-11-26 • 15 questions

03 When Numeracy Hub shows a square-root key, which check confirms a computed positive root?

A. Add the original area

B. Square the result

C. Double the result

D. Divide the result by 2



Numeracy Hub

Choice (A–D)

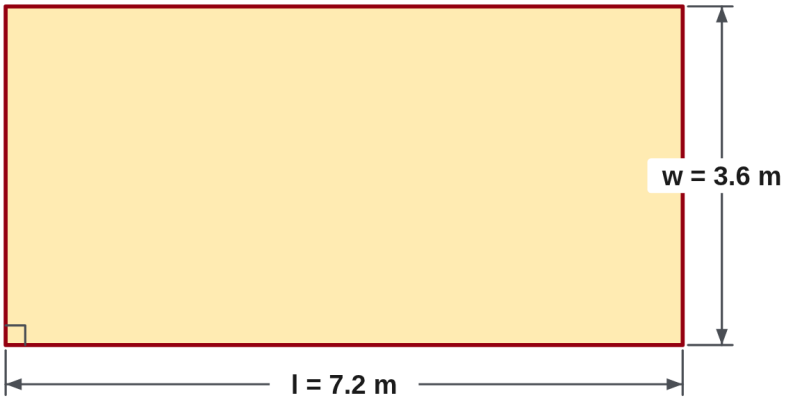
Working / reason

Textbook sections 8.6 pp.201–203; sections 8.7 pp.203–204; calculator pp.51–53

Hidden dimensions: square roots and formula checks

Day 16 • 2026-11-26 • 15 questions

04 A new rectangle keeps the 3.6 m width shown in Shape Canvas but has area 32.4 m². Find its new length.



Shape Canvas

Answer (m)

Working / reason

Textbook sections 8.6 pp.201–203; sections 8.7 pp.203–204; calculator pp.51–53

05 A rectangle has known area and one known side. Which method finds the other side?

A. Always take the square root of area

B. Divide perimeter by 4

C. Square the known side

D. Divide area by the known side

Choice (A–D)

Working / reason

Textbook sections 8.6 pp.201–203; sections 8.7 pp.203–204; calculator pp.51–53

Hidden dimensions: square roots and formula checks

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CALCULATE

06 A square panel has area 14400 mm². Find its side length.

Answer (mm) | **Working / reason**

Textbook sections 8.6 pp.201–203; sections 8.7 pp.203–204; calculator pp.51–53

07 A square floor has area 30.25 m². Find its side length.

Answer (m) | **Working / reason**

Textbook sections 8.6 pp.201–203; sections 8.7 pp.203–204; calculator pp.51–53

08 A square has area 2 m². Find its side to 3 decimal places.

Answer (m) | **Working / reason**

Textbook sections 8.6 pp.201–203; sections 8.7 pp.203–204; calculator pp.51–53

09 A rectangular panel has area 4.32 m² and length 2.4 m. Find its width.

Answer (m) | **Working / reason**

Textbook sections 8.6 pp.201–203; sections 8.7 pp.203–204; calculator pp.51–53

Hidden dimensions: square roots and formula checks

Day 16 • 2026-11-26 • 15 questions

10 A circular area formula is A = pi × r². Use pi = 3.14 and A = 31400 mm² to find r.

Answer (mm)

Working / reason

Textbook sections 8.6 pp.201–203; sections 8.7 pp.203–204; calculator pp.51–53

APPLY & CHECK

11 A student calculates √64 as 32. Which check exposes the error?

A. 32 + 32 = 64

B. 64 ÷ 2 = 32

C. 32 − 32 = 0

D. 32² = 1024, not 64

Choice (A–D)

Working / reason

Textbook sections 8.6 pp.201–203; sections 8.7 pp.203–204; calculator pp.51–53

12 A square’s proposed side is 1.8 m. Calculate its area as a substitution check.

Answer (m²)

Working / reason

Textbook sections 8.6 pp.201–203; sections 8.7 pp.203–204; calculator pp.51–53

Hidden dimensions: square roots and formula checks

Day 16 • 2026-11-26 • 15 questions

13 For r = √(A/pi), which calculator grouping is correct?

A. Multiply √A by pi

B. Take the square root of the whole quotient A/pi

C. Take √A, then divide by pi

D. Divide A by √pi, then stop

Choice (A–D)

Working / reason

Textbook sections 8.6 pp.201–203; sections 8.7 pp.203–204; calculator pp.51–53

14 A square area is 9 m². Its side will be recorded in millimetres. Find the side in mm.

Answer (mm)

Working / reason

Textbook sections 8.6 pp.201–203; sections 8.7 pp.203–204; calculator pp.51–53

15 A square-root calculation gives a side of about 4.47 m for area 20 m². Which estimate supports it?

A. The side is between 20 m and 25 m

B. The side must be 10 m

C. The side is less than 1 m

D. The side is between 4 m and 5 m

Choice (A–D)

Working / reason

Textbook sections 8.6 pp.201–203; sections 8.7 pp.203–204; calculator pp.51–53